

Bento-Exoskel-UI Master Skill MD

A specialized Bento UI design system and component architecture engineered for high-contrast exoskeleton interfaces, tactical HUDs, physical telemetry monitors, and industrial hardware controls.

Built around a Dark Base Canvas (#18181b), Light/Beige Layering (#f4f4ee), and Glass Black Tinted Overlays (--exoskel-glass-tinted), this framework merges the refined depth aesthetics of UI8 with production-grade GSAP animation hooks.

1. Core Stack & External Reference Libraries

The implementation techniques and performance optimizations in this framework are derived from and built upon the following developer tools and documentation:

Tailwind CSS: Utility-first styling framework

used for structural layout, CSS variables, backdrop blurs, and rapid micro-spacing.

GSAP (GreenSock Animation Platform):

Professional-grade JavaScript animation engine used for high-performance UI state transitions, timeline sequences, and slide-out drawer dynamics.

UI8 Design Marketplace: Primary visual benchmark for multi-layered depth, claymorphic card structures, and soft-glow lighting effects.

React Framework: Component architecture for managing interactive states (e.g., locked/unlocked drawers, audio source switching, and interactive viewports).

Lucide Icons: Lightweight vector icon library for interface triggers, audio indicators, and lock/unlock file states.

Key Framework Features

This system integrates a dedicated feature

suite tailored specifically for physical telemetry and hardware control:

Multi-Layer Depth System: Combines a #18181b dark foundation canvas, a high-visibility #f4f4ee beige contrast layer, and a floating glass-black tinted overlay (rgba(20, 20, 25, 0.75)) with backdrop blur filters.

Inline DEV Slide-Out Drawer: Uses GSAP to expand inline diagnostic panels directly within the browser view without forcing subdomain redirects or full page reloads.

Actuator Viewport with Scroll Wheel

(Furniture Card 4): A fixed-boundary card embedding a vertical scrolling preset selector inside the far-right viewport boundary for real-time joint torque and resistance tuning.

Inverted Corner & Exploded Lock System: Features CSS clip-path polygon styling

paired with an interactive .exploded-lock-trigger button to toggle hidden feature capability cards.

Universal Feature Stack: A reusable module wrapper providing live-stream audio visualizers, pre-recorded audio library selectors, and command prompt input capability across any card element.

Tactical Pill Micro-Controls: High-tactile rounded pill buttons (rounded-full) engineered for high-contrast visibility and touch/glove activation on field equipment.

Visual Language & Token Architecture

```
:root {  
  /* Layer 1: Dark Canvas Base */  
  --exoskel-bg-base: #18181b;  
  --exoskel-card-dark: #242427;  
  --exoskel-viewport-dark: #1c1c1f;  
  --exoskel-border-dark: rgba(255, 255, 255,  
    0.08);
```

```
/* Layer 2: Light / Beige Layer Variant */  
--exoskel-bg-light: #e5e5e0;  
--exoskel-card-beige: #f4f4ee;  
--exoskel-beige-shadow: 0 20px 40px -15px  
rgba(0, 0, 0, 0.07);
```

```
/* Layer 3: Glass Black Tinted Overlay */  
--exoskel-glass-tinted: rgba(20, 20, 25, 0.75);  
--exoskel-glass-border: rgba(255, 255, 255, 0.12);  
--exoskel-glass-glow: 0 12px 32px rgba(0, 0, 0,  
0.4);
```

```
/* Radii */  
--exoskel-radius-card: 28px;  
--exoskel-radius-inner: 16px;  
--exoskel-radius-pill: 9999px;  
}
```

```
/* Glass Tinted Overlay Class */
```

```
.ui8-glass-overlay {  
background: var(--exoskel-glass-tinted);  
backdrop-filter: blur(20px) saturate(160%);  
-webkit-backdrop-filter: blur(20px)  
saturate(160%);  
border: 1px solid var(--exoskel-glass-border);  
box-shadow: var(--exoskel-glass-glow);  
}
```

/* Inverted Corner Polygon Cutout */

```
.bento-card-inverted-corner {  
clip-path: polygon(  
0 0,  
calc(100% - 32px) 0,  
100% 32px,  
100% 100%,  
0 100%  
);  
}
```

```
/* Exploded Lock Trigger Button */  
.exploded-lock-trigger {  
  position: absolute;  
  top: 6px;  
  right: 6px;  
  width: 32px;  
  height: 32px;  
  border-radius: var(--exoskel-radius-inner);  
  background: rgba(255, 255, 255, 0.08);  
  border: 1px solid rgba(255, 255, 255, 0.15);  
  display: flex;  
  align-items: center;  
  justify-content: center;  
  cursor: pointer;  
  transition: transform 0.3s cubic-bezier(0.34,  
1.56, 0.64, 1);  
}
```

```
.exploded-lock-trigger:hover {  
  transform: scale(1.1) rotate(5deg);
```

}

4. Component Implementations

[4.1](#) Telemetry Waveform Card (Dark Base + Glass Overlay)

```
import React from 'react';
```

```
export function TelemetryShareCard() {  
  return (  
    <div className="w-[360px] p-6 rounded-[28px]  
      bg-[#242427] border border-white/10 flex flex-  
      col justify-between shadow-2xl">  
      <div className="relative w-full h-56 rounded-2xl  
        bg-[#1c1c1f] border border-white/5 overflow-  
        hidden flex items-center justify-center">  
        { /* Grid Lines Overlay */ }  
        <div className="absolute inset-0 grid grid-  
          cols-3 grid-rows-3 opacity-20 border-white/20">  
          <div className="border-r border-b border-white/
```


20" />

<div className="border-r border-b border-white/
20" />

<div className="border-b border-white/20" />

<div className="border-r border-b border-white/
20" />

<div className="border-r border-b border-white/
20" />

<div className="border-b border-white/20" />
</div>

{/* Floating Glass Black Overlay */}

<div className="ui8-glass-overlay relative w-52
p-3 rounded-2xl flex flex-col gap-2 z-10
transform -rotate-1">

<div className="w-full h-20 rounded-xl bg-
black/40 border border-white/10 flex items-
center justify-center px-3">

<div className="flex items-center gap-[3px]
h-8">

```
{[40, 70, 30, 90, 100, 60, 40, 85, 50, 30, 75,  
45].map((h, i) => (  
<span key={i} className="w-[3px] bg-white/80  
rounded-full" style={{ height: ${h}% }} />  
))}  
</div>  
</div>
```

```
<div className="flex items-center justify-  
between">  
<div className="w-10 h-10 rounded-lg bg-white/  
5 border border-white/10 flex items-center  
justify-center text-[10px] text-white/60">
```

AUDIO

```
</div>
```

```
<button className="w-10 h-10 rounded-lg bg-  
white/10 border border-white/20 flex items-  
center justify-center text-white hover:bg-white/  
20">
```



```
</button>
```

```
</div>
```

```
<div className="absolute -bottom-3 -right-6 flex  
items-center gap-1.5 px-3 py-1 bg-[#3a3a40]  
border border-white/20 rounded-full shadow-lg">  
<span className="text-[11px] font-medium text-  
white">Artur</span>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<div className="mt-6 space-y-2">
```

```
<h3 className="text-xl font-bold text-white  
tracking-tight">One-Click File Sharing</h3>
```

```
<p className="text-sm text-white/60 leading-  
relaxed">
```

Simplify sharing with a single click for any file size.

```
</p>
```

```
</div>
```

```
<div className="mt-6">
```

```
<button className="px-6 py-2.5 text-xs font-  
semibold text-white/90 bg-white/10 border  
border-white/10 rounded-full hover:bg-white/15">  
Discover
```

```
</button>
```

```
</div>
```

```
</div>
```

```
);
```

```
}
```

[4.2](#) Actuator Furniture Card (4) with Viewport Scroll Wheel

```
export function ActuatorFurnitureCard4() {  
  return (
```

```
<div className="w-[360px] p-6 rounded-[28px]  
bg-[#242427] border border-white/10 flex flex-  
col justify-between shadow-2xl">
```

```
<div className="flex items-center justify-between mb-4">
<div className="flex items-center gap-2">
<div className="w-2 h-2 rounded-full bg-emerald-400" />
<span className="text-sm font-semibold text-white">Actuator Preset</span>
</div>
<button className="w-8 h-8 rounded-full bg-white/10 border border-white/20 flex items-center justify-center text-white">

</button>
</div>
```

```
{/* Main Viewport Container */}
<div className="relative w-full h-48 rounded-2xl bg-[#1c1c1f] border border-white/10 overflow-hidden flex items-center justify-between p-4">
<div className="flex-1 h-full flex flex-col items-
```

```
center justify-center text-center">
<span className="text-2xl font-bold text-
white">320 Nm</span>
<span className="text-[10px] text-white/50
uppercase tracking-widest">Joint Torque</span>
</div>
```

```
{/* Far-Right Viewport Scroll Wheel */}
<div className="w-12 h-full border-l border-
white/10 pl-2 flex flex-col items-center justify-
center gap-2 overflow-y-scroll no-scrollbar">
{['P1', 'P2', 'P3', 'P4', 'P5'].map((item, idx) => (
<div key={idx} className="w-8 h-8 rounded-lg
bg-white/5 border border-white/10 flex-shrink-0
flex items-center justify-center text-[10px] text-
white/70 hover:bg-white/10 cursor-pointer">
{item}
</div>
))}
</div>
```

```
</div>
```

```
<div className="mt-4 space-y-1">
```

```
<h4 className="text-sm font-semibold text-white">Limb Calibration</h4>
```

```
<p className="text-xs text-white/50">Adjust  
joint response profiles dynamically.</p>
```

```
</div>
```

```
</div>
```

```
);
```

```
}
```

[4.3](#) Inline GSAP DEV Drawer Component

```
import React, { useState, useRef, useEffect }  
from 'react';
```

```
import gsap from 'gsap';
```

```
export function DevExoskelCard() {  
  const [isUnlocked, setIsUnlocked] =  
    useState(false);
```

```
const slidePanelRef =  
useRef<HTMLDivElement>(null);
```

```
useEffect(() => {  
  if (!slidePanelRef.current) return;  
  if (isUnlocked) {  
    gsap.to(slidePanelRef.current, {  
      xPercent: -100,  
      opacity: 1,  
      duration: 0.5,  
      ease: 'power3.out',  
    });  
  } else {  
    gsap.to(slidePanelRef.current, {  
      xPercent: 0,  
      opacity: 0,  
      duration: 0.4,  
      ease: 'power3.in',  
    });  
  }  
}
```



```
}, [isUnlocked]));
```

```
return (
```

```
<div className="relative overflow-hidden w-[360px]">
```

```
<div className="bento-card-inverted-corner w-full p-6 rounded-[28px] bg-[#242427] border border-white/10 shadow-2xl relative">
```

```
<button
```

```
onClick={() => setIsUnlocked(!isUnlocked)}
```

```
className="exploded-lock-trigger"
```

```
title={isUnlocked ? "Lock Panel" : "Unlock DEV Capabilities"}
```

```
>
```

```
{isUnlocked ? '🔓' : '🔒'}
```

```
</button>
```

```
<div className="pr-8 space-y-2">
```

```
<h3 className="text-lg font-bold text-white">DEV Capabilities</h3>
```

```
<p className="text-xs text-white/50">Toggle  
lock to slide out inline diagnostics panel.</p>
```

```
</div>
```

```
</div>
```

```
{/* GSAP Animated Slide-Out Panel */}
```

```
<div
```

```
  ref={slidePanelRef}
```

```
  className="absolute top-0 left-full w-full h-full  
  p-6 rounded-[28px] ui8-glass-overlay flex flex-col  
  justify-between opacity-0"
```

```
>
```

```
<div className="space-y-2">
```

```
<h4 className="text-sm font-bold text-white  
uppercase tracking-wider">Kernel Diagnostics</  
h4>
```

```
<div className="text-[11px] font-mono text-  
emerald-400 bg-black/50 p-3 rounded-xl border  
border-white/10 space-y-1">
```

```
<div>SYS_BUS: OPTIMAL</div>
```

```
<div>ACTUATOR_TEMP: 38.2°C</div>
```

```
<div>LATENCY: 0.42ms</div>
```

```
</div>
```

```
</div>
```

```
<button
```

```
onClick={() => setIsUnlocked(false)}
```

```
className="px-4 py-1.5 text-xs text-white/70 bg-
```

```
white/10 rounded-full border border-white/10
```

```
hover:bg-white/20"
```

```
>
```

```
Close Panel
```

```
</button>
```

```
</div>
```

```
</div>
```

```
);
```

```
}
```

5. Summary Checklist

Component Feature	Primary Surface	Interaction Model
Telemetry Share Card	Dark #242427 + Glass Overlay	Waveform visualization & cursor badges
Actuator Furniture Card (4)	Dark Viewport #1c1c1f	In-viewport vertical scroll wheel
Inline DEV Drawer	Inverted Corner + Glass Overlay	GSAP lock-triggered slide drawer
Power Cell Vault	Light Beige #f4f4ee	Gauge meter & soft contrast control

